

Robotics Software Developer

# JOSEF GSTOETTNER

**ROS/ROS2 | C++ | Python | Gazebo**

## CONTACT


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## SKILLS

### Areas of specialization

- ROS / ROS 2 (Navigation, SLAM, Moveit)
- Robot Simulation in Gazebo & Isaacsim
- Behavior trees, robot task management
- Embedded software (STM32, Arduino)
- Single board computers (Raspberry Pi, Jetson)

### Programming

Python
C / C++
Docker
C#
Matlab
JavaScript
HTML, CSS

OS:

### Languages

| | |
|---|---|
| **English** | Full Professional |
| **Mandarin** | Limited Working |
| **German** | Native |

### Learning & Hobbies

- PCB design with KiCAD
- Android development

C++ and Python developer specialized in ROS 2, navigation, SLAM and behavior trees for reliable autonomous mobile robots. Have a strong understanding of mechanical engineering principles, object-oriented programming and robot dynamics to optimize system performance.

## Work History

**Senior Robotics Software Engineer** 2026/3–present
ROS 2 ROBOT PROGRAMMING · Primech Ai | Sciences Park
- Set up servo motor control, navigation & SLAM for mecanum wheeled agv.
- Packaged robot software and tools with Docker for easy deployment.
- Implement Open-RMF to manage an autonomous fleet of toilet-cleaning robots.
- Setup CI/CD for automatic testing of the whole ROS2 software and building the docker image.
- Configure Seer-roboshop navigation & SLAM for mecanum wheeled agv.

**Assistant Research Engineer** 2025/9–2026/3
ROS 2 ROBOT PROGRAMMING · HKCRC | Sciences Park
- Develop autonomous tasks and recovery behaviors for a water tank cleaning robot using behavior trees.
- Set up navigation, SLAM & coverage planner to fully clean the whole surface.
- Packaged robot software and tools with Docker for easy deployment.
- Create realistic Gazebo simulations for faster iterations and testing.

**System Engineer** 2022/4–2024/8
ROS 2 ROBOT PROGRAMMING · LSCM | Sciences Park
- Set up navigate & SLAM (Cartographer, SlamToolbox, ...) for 2D/3D mapping of large areas.
- Simulated robots in Gazebo for testing and faster development.
- Developed autonomous docking utilizing AprilTags for precise movement.
- Tuned PID controllers for smooth movement and speed control.

**Research Assistant** 2020/7–2022/3
EMBEDDED SOFTWARE · HKUST | Clear Water Bay
- CAD design for 3D printing and laser cutting of prototypes.
- Firmware development on a low-power IoT accelerometer with BLE Mesh for predictive maintenance based on Nrf52.

**Mechanical Engineer** 2019/8–2020/4
CAD DESIGN · KALBAS | Cheung Sha Wan

## Education

**Master of Science: Mechanical Engineering** 2018/9–2019/6
· HKUST | Clear Water Bay

**One Year Mandarin language study** 2017/9–2018/6
· 西南交通大学 | 四川峨嵋 | 汉语水平考试四级

**Bachelor of Science: Mechanical Engineering** 2014/9–2017/6
· University of Applied Sciences Upper Austria | Austria

## Projects

**Robomaster** 2021/10–2022/4
SOFTWARE TEAM · HKUST
- SLAM and navigation for autonomous Robots in ROS
- Embedded software development on STM32.

**Autonomous RC-car race (first place 🏆)** 2020/12–2021/3
IMITATION LEARNING · HKUST https://github.com/JosefGst/autorace
- Trained Pytorch model to be used on Jetson Nano for autonomous-driving, obstacle avoidance and overtaking of other cars.